

ENME608 Mechanisms

Force Analysis — Equation of Motion (Norton Ch. 11)

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Contents

1	Newtonian Solution Method (Free-Body Diagram Approach)	2
2	Single Link in Pure Rotation	2
3	Four-Bar Linkage — Matrix Method	3
3.1	Free-Body Diagrams and Unknowns	3
3.2	Equations of Motion per Link	3
3.3	Full 9×9 Matrix	3
4	Crank-Slider — Matrix Method	3
5	Virtual Work / Energy Method	4
6	Shaking Force and Shaking Moment	4
7	Flywheel Design (Passive Torque Smoothing)	5
7.1	Concept	5
7.2	Energy Method	5
7.3	Sizing the Flywheel	5
8	Tutorial Solutions	5
8.1	P11-5 — Four-Bar Linkage, Matrix Method (Row a)	5
8.2	P11-6 — Four-Bar Linkage, Virtual Work Method (Row a)	7
8.3	P11-3 — Crank-Slider, Matrix Method (Row a)	8
8.4	P11-4 — Crank-Slider, Virtual Work Method (Row a)	9

Newtonian Solution Method (Free-Body Diagram Approach)

Force analysis determines the joint reaction forces and the driving torque required in a mechanism, given the kinematics and inertial properties of each link.

Governing equations (2-D, per link):

$$\sum F_x = m a_{Gx}, \quad \sum F_y = m a_{Gy}, \quad \sum T_G = I_G \alpha$$

Sign convention and notation:

- F_{ij} = force exerted *on* link j *by* link i . Newton's 3rd: $F_{ji} = -F_{ij}$.
- $\mathbf{R}_{ij}$ = position vector from the CG of link j to the point where F_{ij} is applied.
- Torque from a force (2-D scalar, $\hat{k}$ component only):

$$T = R_x F_y - R_y F_x$$

Coordinate systems:

- **GCS** — Global Coordinate System: fixed X, Y .
- **LNCS** — Local Non-rotating Coordinate System: origin at link CG, axes parallel to X, Y , does not rotate with link.
- **LRCS** — Local Rotating Coordinate System: origin at link CG, x -axis along link, rotates with link.

Exam focus. Gravity is **neglected** when COG accelerations $\gg g = 386 \text{ in/s}^2$ (typical for high-speed mechanisms). If gravity must be included, treat $m\mathbf{g}$ as an additional external force on each link. Default units throughout are **IPS**: lb, in, s, blob ($= \text{lb} \cdot \text{s}^2/\text{in}$).

Single Link in Pure Rotation

For a single link (link 2) pinned to ground at one end with an applied external force $\mathbf{F}_P$ at point P :

Unknowns: F_{12x}, F_{12y}, T_{12} (3 unknowns)

Equations: 3 scalar EOMs $\Rightarrow 3 \times 3$ linear system

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -R_{12y} & R_{12x} & 1 \end{bmatrix} \begin{bmatrix} F_{12x} \\ F_{12y} \\ T_{12} \end{bmatrix} = \begin{bmatrix} m_2 a_{Gx} - F_{Px} \\ m_2 a_{Gy} - F_{Py} \\ I_{G2} \alpha_2 - (R_{Px} F_{Py} - R_{Py} F_{Px}) \end{bmatrix}$$

where $\mathbf{R}_{12}$ points from the CG of link 2 to pin 12 (the ground pivot), and $\mathbf{R}_P$ points from the CG to the point of application of F_P .

Exam focus. The right-hand side is **inertia minus applied**: ma terms go on the RHS with the externals subtracted. Sign errors here are the most common mistake.

Four-Bar Linkage — Matrix Method

Free-Body Diagrams and Unknowns

Unknowns: $F_{12x}, F_{12y}, F_{32x}, F_{32y}, F_{43x}, F_{43y}, F_{14x}, F_{14y}, T_{12}$
 $\Rightarrow 9$ unknowns, 9×9 system

Equations of Motion per Link

Link 2 (crank):

$$\begin{aligned} F_{12x} + F_{32x} &= m_2 a_{G2x}, & F_{12y} + F_{32y} &= m_2 a_{G2y} \\ (-R_{12y} F_{12x} + R_{12x} F_{12y}) &+ (-R_{32y} F_{32x} + R_{32x} F_{32y}) + T_{12} &= I_{G2} \alpha_2 \end{aligned}$$

Link 3 (coupler), with external force F_P at R_P and external torque T_3 :

$$\begin{aligned} F_{23x} + F_{43x} + F_{Px} &= m_3 a_{G3x}, & F_{23y} + F_{43y} + F_{Py} &= m_3 a_{G3y} \\ (-R_{23y} F_{23x} + R_{23x} F_{23y}) &+ (-R_{43y} F_{43x} + R_{43x} F_{43y}) + (R_{Px} F_{Py} - R_{Py} F_{Px}) + T_3 &= I_{G3} \alpha_3 \end{aligned}$$

where $F_{23x} = -F_{32x}$, $F_{23y} = -F_{32y}$ (Newton's 3rd).

Link 4 (rocker), with external torque T_4 :

$$\begin{aligned} F_{34x} + F_{14x} &= m_4 a_{G4x}, & F_{34y} + F_{14y} &= m_4 a_{G4y} \\ (-R_{34y} F_{34x} + R_{34x} F_{34y}) &+ (-R_{14y} F_{14x} + R_{14x} F_{14y}) + T_4 &= I_{G4} \alpha_4 \end{aligned}$$

where $F_{34x} = -F_{43x}$, $F_{34y} = -F_{43y}$.

Full 9×9 Matrix

Ordering unknowns as $\mathbf{b} = [F_{12x}, F_{12y}, F_{32x}, F_{32y}, F_{43x}, F_{43y}, F_{14x}, F_{14y}, T_{12}]^T$:

$$[A]_{9 \times 9} \mathbf{b} = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ -R_{12y} & R_{12x} & -R_{32y} & R_{32x} & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & R_{32y} & -R_{32x} & -R_{43y} & R_{43x} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & R_{43y} & -R_{43x} & -R_{14y} & R_{14x} & 0 \end{bmatrix} \begin{bmatrix} F_{12x} \\ F_{12y} \\ F_{32x} \\ F_{32y} \\ F_{43x} \\ F_{43y} \\ F_{14x} \\ F_{14y} \\ T_{12} \end{bmatrix} = \begin{bmatrix} m_2 a_{G2x} \\ m_2 a_{G2y} \\ I_{G2} \alpha_2 \\ m_3 a_{G3x} - F_{Px} \\ m_3 a_{G3y} - F_{Py} \\ I_{G3} \alpha_3 - (R_{Px} F_{Py} - R_{Py} F_{Px}) - T_3 \\ m_4 a_{G4x} \\ m_4 a_{G4y} \\ I_{G4} \alpha_4 - T_4 \end{bmatrix}$$

Exam focus. The exam will NOT ask you to invert the 9×9 matrix by hand. You must be able to:
 (1) correctly identify all $\mathbf{R}$ vectors (from CG to pin, in global x, y); (2) write the correct RHS terms;
 (3) set up the matrix structure.

Crank-Slider — Matrix Method

The slider (link 4) can only exert a force perpendicular to the slide rail (F_{14y}), so F_{14x} is replaced by a friction constraint:

$$F_{14x} = -\mu \cdot \text{SGN}(\dot{d}) \cdot |F_{14y}|$$

where μ is the coefficient of friction and $\dot{d}$ is the slider velocity.

Unknowns: $F_{12x}, F_{12y}, F_{32x}, F_{32y}, F_{43x}, F_{43y}, F_{14y}, T_{12}$
 $\Rightarrow 8$ unknowns, 8×8 system

Link 4 equations (frictionless slide, $\mu = 0$, so $F_{14x} = 0$):

$$F_{34x} = m_4 a_{G4x}, \quad F_{34y} + F_{14y} = m_4 a_{G4y}, \quad -R_{34y} F_{34x} + R_{34x} F_{34y} = I_{G4} \alpha_4$$

For the frictionless case $\alpha_4 = 0$ (slider translates only) and I_{G4} is irrelevant. The moment equation for link 4 collapses to just F_{43} components.

Exam focus. F_{14x} is never an unknown in the crank-slider; it is either zero (frictionless) or expressed through the friction constraint. Always label which direction the slider moves to get the correct sign of μ .

Virtual Work / Energy Method

The virtual work method yields T_{12} from a **single scalar equation** without solving for internal joint forces. All internal constraint forces do no virtual work (they are equal and opposite at the same point).

$$\sum_k \mathbf{F}_k \cdot \mathbf{V}_{Gk} + \sum_k T_k \omega_k - \sum_k m_k \mathbf{a}_{Gk} \cdot \mathbf{V}_{Gk} - \sum_k I_{Gk} \alpha_k \omega_k = 0$$

Solving for T_{12} :

$$T_{12} = \frac{1}{\omega_2} \left[\sum_{k \neq \text{input}} m_k \mathbf{a}_{Gk} \cdot \mathbf{V}_{Gk} + \sum_k I_{Gk} \alpha_k \omega_k - \sum_k \mathbf{F}_{P,k} \cdot \mathbf{V}_{P,k} - \sum_{k \neq \text{input}} T_k \omega_k \right]$$

where $\mathbf{F}_{P,k}$ and $\mathbf{V}_{P,k}$ are the applied external forces and velocities at the point of application on link k .

Dot product in component form:

$$\mathbf{a}_G \cdot \mathbf{V}_G = a_{Gx} V_{Gx} + a_{Gy} V_{Gy}$$

Exam focus. Virtual work gives **only** T_{12} . To find joint forces you still need the FBD matrix method. The two methods should give identical T_{12} values — a useful cross-check.

Shaking Force and Shaking Moment

The net force and moment transmitted to the **ground** (frame, link 1) by the mechanism:

$$\mathbf{F}_s = \mathbf{F}_{21} + \mathbf{F}_{41} \quad (\text{sum of all ground-pin reaction forces})$$

$$M_s = T_{21} + (\mathbf{R}_1 \times \mathbf{F}_{41})_z$$

where $\mathbf{R}_1$ is the vector from the reference point (e.g. pin 12) to pin 14, and $T_{21} = -T_{12}$ (reaction on ground from input shaft).

Shaking force/moment cause vibration and noise in the machine frame; balancing aims to minimise them.

Flywheel Design (Passive Torque Smoothing)

Concept

A flywheel stores/releases kinetic energy to smooth out a periodically varying load or driver torque. The motor torque T_M is set equal to the average value T_{avg} ; excess/deficit energy is exchanged with the flywheel.

Energy Method

$$T_M - T_L(\theta) = I_s \alpha, \quad (T_{\text{avg}} - T_L) d\theta = I_s \omega d\omega$$

Integrating over one cycle gives the energy fluctuation:

$$E = \int_{\theta_1}^{\theta_2} (T_{\text{avg}} - T_L) d\theta = \frac{1}{2} I_s (\omega_{\text{max}}^2 - \omega_{\text{min}}^2)$$

Sizing the Flywheel

Coefficient of fluctuation:

$$k = \frac{\omega_{\text{max}} - \omega_{\text{min}}}{\omega_{\text{avg}}}$$

Required system MOI:

$$I_s = \frac{E}{k \omega_{\text{avg}}^2}$$

Flywheel MOI:

$$I_f = I_s - I_{\text{other shaft components}}$$

Flywheel radius (treating as thin ring of mass m):

$$I_f = mr^2 \quad \Rightarrow \quad r = \sqrt{I_f/m}$$

Exam focus. Steps: (1) plot T_L vs θ ; (2) find T_{avg} (area/ 2π); (3) identify max positive and negative area segments to get E ; (4) apply $I_s = E/(k\omega_{\text{avg}}^2)$; (5) subtract other rotating inertias to get I_f ; (6) solve for r .

Tutorial Solutions

P11-5 — Four-Bar Linkage, Matrix Method (Row a)

Given:

- Links: $l_2 = 4$, $l_3 = 12$, $l_4 = 8$, $l_1 = 15$ in (ground)
- $\theta_2 = 45$, $\theta_3 = 24.97$, $\theta_4 = 99.30$
- $\omega_2 = 20$, $\omega_3 = -5.62$, $\omega_4 = 3.56$ rad/s
- $\alpha_2 = 20$, $\alpha_3 = 75.29$, $\alpha_4 = 244.43$ rad/s²
- $m_2 = 0.002$, $m_3 = 0.020$, $m_4 = 0.100$ blobs

- $I_{G2} = 0.10, I_{G3} = 0.20, I_{G4} = 0.50 \text{ blob}\cdot\text{in}^2$
- $T_3 = -15 \text{ lb}\cdot\text{in}, T_4 = 25 \text{ lb}\cdot\text{in}$
- $R_{g2} = 2@0$ from pin (CG on link 2), $R_{g3} = 5@0, R_{g4} = 4@30$
- $F_{P3} = 0; F_{P4} = 40@-30 \text{ lb}$ applied at $R_{P4} = 8@0$ from CG of link 4
- $a_{G2} = 801.00@222.14, a_{G3} = 1691.49@208.24, a_{G4} = 979.02@222.27 \text{ in/s}^2$

Step 1 — Position Vectors (from CG to pin, global x,y)

Each $\mathbf{R}$ vector is computed in global x, y components from the kinematic configuration.

Link 2: $R_{g2} = 2$, CG lies 2 in from pin 12 along link 2 ($\theta_2 = 45$), so CG is at $(2 \cos 45, 2 \sin 45) = (1.414, 1.414)$ from pin 12. Vector from CG to pin 12: $\mathbf{R}_{12} = (-1.414, -1.414) \text{ in}$. Vector from CG to pin 32: the far end of link 2 is at $4 \cos 45, 4 \sin 45$ from pin 12, so from CG: $\mathbf{R}_{32} = (4 - 2) \cos 45, (4 - 2) \sin 45 = (1.414, 1.414) \text{ in}$.

Link 3: $R_{g3} = 5$ along link 3 ($\theta_3 = 24.97$):

$$\mathbf{R}_{23} = -5(\cos 24.97, \sin 24.97) = (-4.533, -2.110) \text{ in}$$

$$\mathbf{R}_{43} = (12 - 5)(\cos 24.97, \sin 24.97) = (6.346, 2.955) \text{ in}$$

Link 4: $R_{g4} = 4@30$ relative to link 4 ($\theta_4 = 99.30$), i.e. the CG is at angle $\theta_4 + 30 = 129.30$ from pin 14 measured in global frame:

$$\mathbf{R}_{14} = -4(\cos 129.30, \sin 129.30) = (2.534, -3.095) \text{ in (from CG to pin 14)}$$

$$\mathbf{R}_{34} = (8 - 4)(\cos 99.30, \sin 99.30) \approx \dots = (1.241, 4.800) \text{ in (from CG to pin 34)}$$

Applied force point on link 4 ($R_{P4} = 8@0$ from pin 14, measured along link): At angle $\theta_4 = 99.30$ from pin 14, end of link 4 is $8(\cos 99.30, \sin 99.30)$. CG is $4@(99.30 + 30)$ from pin 14.

$$\mathbf{R}_{P4} = (-1.293, 7.900) \text{ in (from CG to applied force point)}$$

$$F_{P4} = 40@-30 \Rightarrow (F_{P4x}, F_{P4y}) = (34.64, -20.00) \text{ lb.}$$

Step 2 — Acceleration Components

$$\mathbf{a}_{G2} = 801.00\angle 222.14 \Rightarrow (-593.9, -537.4) \text{ in/s}^2$$

$$\mathbf{a}_{G3} = 1691.49\angle 208.24 \Rightarrow (-1490.0, -800.4) \text{ in/s}^2$$

$$\mathbf{a}_{G4} = 979.02\angle 222.27 \Rightarrow (-724.4, -658.5) \text{ in/s}^2$$

Step 3 — Right-Hand Side Vector

$$C_1 = m_2 a_{G2x} = 0.002(-593.9) = -1.188 \text{ lb}$$

$$C_2 = m_2 a_{G2y} = 0.002(-537.4) = -1.075 \text{ lb}$$

$$C_3 = I_{G2} \alpha_2 = 0.10(20) = 2.000 \text{ lb in}$$

$$C_4 = m_3 a_{G3x} - F_{P3x} = 0.020(-1490.0) - 0 = -29.80 \text{ lb}$$

$$C_5 = m_3 a_{G3y} - F_{P3y} = 0.020(-800.4) - 0 = -16.01 \text{ lb}$$

$$C_6 = I_{G3} \alpha_3 - T_3 = 0.20(75.29) - (-15) = 15.058 + 15 = 30.06 \text{ lb in}$$

$$\begin{aligned}
C_7 &= m_4 a_{G4x} - F_{P4x} = 0.100(-724.4) - 34.64 = -107.08 \text{ lb} \\
C_8 &= m_4 a_{G4y} - F_{P4y} = 0.100(-658.5) - (-20.00) = -45.85 \text{ lb} \\
C_9 &= I_{G4} \alpha_4 - T_4 - (R_{P4x} F_{P4y} - R_{P4y} F_{P4x}) \\
C_9 &= 0.50(244.43) - 25 - [(-1.293)(-20.00) - (7.900)(34.64)] \\
&= 122.22 - 25 - [25.86 - 273.66] \\
&= 97.22 + 247.80 = 345.02 \text{ lb in}
\end{aligned}$$

Step 4 — Assemble and Solve

The 9×9 matrix $[A]$ is populated with the $\mathbf{R}$ components as in Section 3.2, right-hand side is $\mathbf{C}$ above. Solving $[A]\mathbf{b} = \mathbf{C}$ yields the nine joint forces and T_{12} .

Exam focus. On the exam: set up the RHS (steps 1–3) and write out the matrix structure. A calculator/MATLAB solves the system; no hand-inversion required.

P11-6 — Four-Bar Linkage, Virtual Work Method (Row a)

Same four-bar as P11-5. Additional kinematic data (Row a):

- $\omega_2 = 20$, $\omega_3 = -5.62$, $\omega_4 = 3.56$ rad/s
- $V_{G2} = 40 \angle 135 \Rightarrow (-28.284, +28.284)$ in/s
- $V_{G3} = 54.44 \angle 145.19 \Rightarrow (-44.70, +31.07)$ in/s
- $V_{G4} = 14.23 \angle 219.30 \Rightarrow (-11.01, -9.01)$ in/s
- $V_{P3} = 54.44 \angle 145.19 \Rightarrow (-44.70, +31.07)$ in/s (same as V_{G3} here)
- $V_{P4} = 41.39 \angle -160.80 \Rightarrow (-38.90, -14.16)$ in/s
- $F_{P3} = 0$; $F_{P4} = (34.64, -20.00)$ lb (from P11-5)

Virtual Work Equation

Applying Eq. 11.16c:

$$\begin{aligned}
T_{12} \omega_2 &= m_2 \mathbf{a}_{G2} \cdot \mathbf{V}_{G2} + m_3 \mathbf{a}_{G3} \cdot \mathbf{V}_{G3} + m_4 \mathbf{a}_{G4} \cdot \mathbf{V}_{G4} \\
&\quad + I_{G2} \alpha_2 \omega_2 + I_{G3} \alpha_3 \omega_3 + I_{G4} \alpha_4 \omega_4 - \mathbf{F}_{P4} \cdot \mathbf{V}_{P4} - T_3 \omega_3 - T_4 \omega_4
\end{aligned}$$

Term-by-Term Evaluation

$$\begin{aligned}
m_2 \mathbf{a}_{G2} \cdot \mathbf{V}_{G2} &= 0.002[(-593.9)(-28.284) + (-537.4)(28.284)] \\
&= 0.002[16800 - 15203] = 0.002(1597) = 3.19 \\
m_3 \mathbf{a}_{G3} \cdot \mathbf{V}_{G3} &= 0.020[(-1490.0)(-44.70) + (-800.4)(31.07)] \\
&= 0.020[66603 - 24868] = 0.020(41735) = 834.7 \\
m_4 \mathbf{a}_{G4} \cdot \mathbf{V}_{G4} &= 0.100[(-724.4)(-11.01) + (-658.5)(-9.01)] \\
&= 0.100[7977 + 5931] = 0.100(13908) = 1390.8 \\
I_{G2} \alpha_2 \omega_2 &= 0.10(20)(20) = 40.0 \\
I_{G3} \alpha_3 \omega_3 &= 0.20(75.29)(-5.62) = -84.7
\end{aligned}$$

$$I_{G4}\alpha_4\omega_4 = 0.50(244.43)(3.56) = 434.9$$

$$\mathbf{F}_{P4} \cdot \mathbf{V}_{P4} = (34.64)(-38.90) + (-20.00)(-14.16) = -1347.5 + 283.2 = -1064.3$$

$$T_3\omega_3 = (-15)(-5.62) = 84.3$$

$$T_4\omega_4 = 25(3.56) = 89.0$$

Substituting:

$$T_{12}(20) = 3.19 + 834.7 + 1390.8 + 40.0 - 84.7 + 434.9 - (-1064.3) - 84.3 - 89.0$$

$$T_{12}(20) = 3509.9 \quad \Rightarrow \quad \boxed{T_{12} \approx 175.5 \text{ lb in}}$$

Exam focus. The virtual work result for T_{12} should agree with the matrix method to within rounding. If they differ substantially, check velocity/acceleration signs and dot-product arithmetic.

P11-3 — Crank-Slider, Matrix Method (Row a)

Given:

- Links: $l_2 = 4$, $l_3 = 12$, offset = 0 in
- $\theta_2 = 45$, $\omega_2 = 10$, $\alpha_2 = 20 \text{ rad/s}$ (rad/s^2)
- $m_2 = 0.002$, $m_3 = 0.020$, $m_4 = 0.060$ blobs
- $I_{G2} = 0.10$, $I_{G3} = 0.20 \text{ blob}\cdot\text{in}^2$
- $R_{g2} = 2@0$, $R_{g3} = 5@0$
- $F_{P3} = 0$; $T_3 = 20 \text{ lb}\cdot\text{in}$ (external torque on coupler)
- $\theta_3 = 166.40$, $\alpha_3 = -2.40 \text{ rad/s}^2$
- $a_{G2} = 203.96@213.69$, $a_{G3} = 371.08@200.84$, $a_{G4} = 357.17@180 \text{ in/s}^2$

Step 1 — Position Vectors

Link 2 ($\theta_2 = 45$, CG at $R_{g2} = 2$ from pin 12):

$$\mathbf{R}_{12} = -2(\cos 45, \sin 45) = (-1.414, -1.414) \text{ in}$$

$$\mathbf{R}_{32} = (4 - 2)(\cos 45, \sin 45) = (1.414, 1.414) \text{ in}$$

Link 3 ($\theta_3 = 166.40$, CG at $R_{g3} = 5$ from pin 23):

$$\mathbf{R}_{23} = -5(\cos 166.40, \sin 166.40) = -5(-0.972, 0.235) = (4.860, -1.176) \dots$$

Wait, sign: $\mathbf{R}_{23}$ is from CG of link 3 to pin 23, so:

$$\mathbf{R}_{23} = -5(\cos 166.40, \sin 166.40) = (-4.860, +1.176) \text{ in (from answer sheet)}$$

$$\mathbf{R}_{43} = (12 - 5)(\cos 166.40, \sin 166.40) = (6.804, -1.646) \text{ in}$$

Note: link 4 is the slider; there is no pin 43 in the rotational sense — F_{43} is the force from the slider on the coupler, applied at the pin at end of link 3. $\mathbf{R}_{43}$ points from CG of link 3 to the slider pin.

Step 2 — Acceleration Components

$$\mathbf{a}_{G2} = 203.96\angle 213.69 \Rightarrow (-169.7, -113.1) \text{ in/s}^2$$

$$\mathbf{a}_{G3} = 371.08\angle 200.84 \Rightarrow (-346.8, -132.0) \text{ in/s}^2$$

$$\mathbf{a}_{G4} = 357.17\angle 180 \Rightarrow (-357.2, 0) \text{ in/s}^2$$

Step 3 — Right-Hand Side (8 equations)

The crank-slider 8×8 system (Eq. 11.10g) follows the same structure as the four-bar except F_{14x} is removed and link 4 has only translational DOF in x :

$$C_1 = m_2 a_{G2x} = 0.002(-169.7) = -0.339 \text{ lb}$$

$$C_2 = m_2 a_{G2y} = 0.002(-113.1) = -0.226 \text{ lb}$$

$$C_3 = I_{G2} \alpha_2 = 0.10(20) = 2.00 \text{ lb in}$$

$$C_4 = m_3 a_{G3x} = 0.020(-346.8) = -6.936 \text{ lb}$$

$$C_5 = m_3 a_{G3y} = 0.020(-132.0) = -2.640 \text{ lb}$$

$$C_6 = I_{G3} \alpha_3 - T_3 = 0.20(-2.40) - 20 = -0.48 - 20 = -20.48 \text{ lb in}$$

$$C_7 = m_4 a_{G4x} = 0.060(-357.2) = -21.43 \text{ lb}$$

$$C_8 = m_4 a_{G4y} = 0.060(0) = 0 \text{ lb}$$

Assemble $[A]_{8 \times 8} \mathbf{b} = \mathbf{C}$ with $\mathbf{b} = [F_{12x}, F_{12y}, F_{32x}, F_{32y}, F_{43x}, F_{43y}, F_{14y}, T_{12}]^T$ and solve.

P11-4 — Crank-Slider, Virtual Work Method (Row a)

Same crank-slider as P11-3. Additional kinematic data:

- $\omega_2 = 10, \omega_3 = -2.43 \text{ rad/s}$ (slider has $\omega_4 = 0$)
- $V_{G2} = 20\angle 135 \Rightarrow (-14.142, +14.142) \text{ in/s}$
- $V_{G3} = 35.24\angle 152.09 \Rightarrow (-31.141, +16.495) \text{ in/s}$
- $V_{G4} = 35.14\angle 180 \Rightarrow (-35.14, 0) \text{ in/s}$
- $V_{P3} = 35.24\angle 152.09 \Rightarrow (-31.127, +16.522) \text{ in/s}$
- $F_{P3} = 0$

Virtual Work Equation

For the crank-slider with $\alpha_4 = 0, \omega_4 = 0, T_3 = 20 \text{ lb}\cdot\text{in}$ (Eq. 11.16c):

$$T_{12} \omega_2 = m_2 \mathbf{a}_{G2} \cdot \mathbf{V}_{G2} + m_3 \mathbf{a}_{G3} \cdot \mathbf{V}_{G3} + m_4 \mathbf{a}_{G4} \cdot \mathbf{V}_{G4} + I_{G2} \alpha_2 \omega_2 + I_{G3} \alpha_3 \omega_3 - T_3 \omega_3$$

Term-by-Term Evaluation

$$\begin{aligned}m_2 \mathbf{a}_{G2} \cdot \mathbf{V}_{G2} &= 0.002[(-169.7)(-14.142) + (-113.1)(14.142)] \\ &= 0.002[2400 - 1599] = 0.002(801) = 1.60\end{aligned}$$

$$\begin{aligned}m_3 \mathbf{a}_{G3} \cdot \mathbf{V}_{G3} &= 0.020[(-346.8)(-31.141) + (-132.0)(16.495)] \\ &= 0.020[10799 - 2177] = 0.020(8622) = 172.4\end{aligned}$$

$$m_4 \mathbf{a}_{G4} \cdot \mathbf{V}_{G4} = 0.060[(-357.2)(-35.14) + (0)(0)] = 0.060(12551) = 753.1$$

$$I_{G2} \alpha_2 \omega_2 = 0.10(20)(10) = 20.0$$

$$I_{G3} \alpha_3 \omega_3 = 0.20(-2.40)(-2.43) = 1.17$$

$$T_3 \omega_3 = 20(-2.43) = -48.6$$

Substituting:

$$T_{12}(10) = 1.60 + 172.4 + 753.1 + 20.0 + 1.17 - (-48.6)$$

$$T_{12}(10) = 996.9 \quad \Rightarrow \quad \boxed{T_{12} \approx 99.7 \text{ lb in}}$$

Exam focus. Always check signs on $T_k \omega_k$ terms: an external torque that *opposes* the mechanism motion contributes *negative* power (positive resistance), which *increases* the required T_{12} . Here $T_3 \omega_3$ is subtracted because it is on the RHS as work done against external torques.